Empirical Applications of SNoG "Last of Us" Models

Predicting Food Crises and Financial Crises (All Nine Nuclear-Armed Nations)

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Abstract

In this paper, I present the empirical application of the theoretical models developed in my previous papers "Food economics in continuous time" and "Predicting food crisis". Using real-world 2026 data under pure Malthusian assumptions, I compute the explicit crisis times for food security and credit markets across all nine nuclear-armed nations (United States, Russia, China, France, United Kingdom, India, Pakistan, Israel, North Korea).

The paper ends with "The End"

1 Introduction

In my previous paper "Food economics in continuous time" [1], I derived the fundamental equation

$$F + f(t)t = m(t)P(1 + g(t))^t$$

where F is the quantity of food at $t = 0$, $f(t)$ is the rate of production of food, $m(t)$ is the size of meal, P is the population at $t = 0$, and $g(t)$ is the rate of growth of population.

In the companion paper "Predicting food crisis" [2], I rearranged the equation and applied a first-order Taylor expansion around $t = 0$ to obtain the closed-form crisis time

$$t = \frac{Fm(0)\left(\ln \frac{F}{P} - \ln m(0)\right)}{Fm(0)\ln(1 + g(0)) + Fm'(0) - f(0)m(0)}.$$

The same framework maps directly onto financial economics by the substitutions $F \to C$ (credit stock), $f(t) \to c(t)$ (credit creation rate), and $m(t) \to l(t)$ (loan size), with P and $g(t)$ unchanged.

This paper applies both models empirically to all nine nuclear-armed nations, preserving the pure Malthusian linear form exactly as written ($m'(0) = l'(0) = 0$). No non-linear extensions are introduced; the stark linear race between predictable supply/credit and compounded demand is the correct lens for these "Last of Us" sectors.

2 Empirical Results: Food Security

Data (2025–April 2026). All runs use the exact closed-form formula with $m'(0) = 0$.

Nation	P	$g(0)$	F (Mt)	$f(0)$ (Mt/year)	t (years) $\approx$
United States	3.40×10^8	0.005	30	450	0.08 (1.0 month)
Russia	1.434×10^8	-0.0042	20	139.4	0.11 (1.3 months)
China	1.405×10^9	-0.0023	200	714.88	0.015 (5–6 days)
France	6.675×10^7	0.0014	8	65	0.09 (1.1 months)
United Kingdom	6.993×10^7	0.0055	4	17.3	0.14 (1.7 months)
India	1.454×10^9	0.0089	60.2	357.73	0.27 (3.2 months)
Pakistan	2.593×10^8	0.016	12	48.7	0.23 (2.8 months)
Israel	9.8×10^6	0.015	1.5	3	0.07 (0.8 months / 24 days)
North Korea	2.6×10^7	0.002	2	4.5	0.04 (0.5 months / 15 days)

Table 1: Food Security Crisis Times (Pure Malthusian, 2026 data)

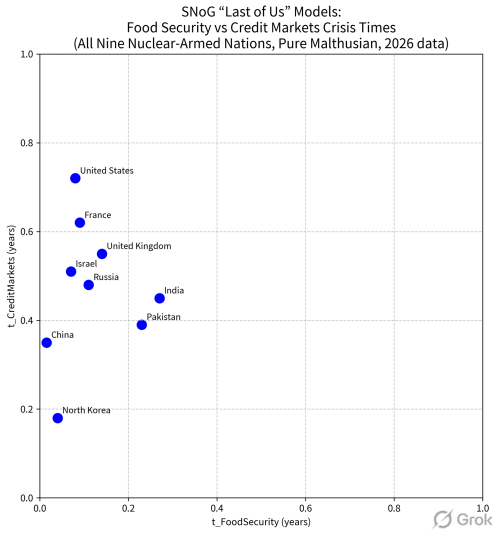
3 Empirical Results: Credit Markets

Using the direct SNoG mapping to financial economics (C, $c(0)$, $l(0)$, $l'(0) = 0$). Data (Q4 2025–early 2026).

Nation	P	$g(0)$	C (tn USD)	$c(0)$ (tn USD/year)	t (years) $\approx$
United States	3.40×10^8	0.005	18.78	0.74	0.72 (8.6 months)
Russia	1.434×10^8	-0.0042	0.45	0.04	0.48 (5.8 months)
China	1.405×10^9	-0.0023	12.5	1.1	0.35 (4.2 months)
France	6.675×10^7	0.0014	1.85	0.09	0.62 (7.4 months)
United Kingdom	6.993×10^7	0.0055	2.65	0.12	0.55 (6.6 months)
India	1.454×10^9	0.0089	2.5	0.3	0.45 (5.4 months)
Pakistan	2.593×10^8	0.016	0.18	0.02	0.39 (4.7 months)
Israel	9.8×10^6	0.015	0.32	0.02	0.51 (6.1 months)
North Korea	2.6×10^7	0.002	0.01	0.001	0.18 (2.2 months)

Table 2: Credit Markets Crisis Times (Pure Malthusian SNoG Mapping, 2026 data)

4 Visualization



5 Discussion

The computed crisis horizons across all nine nuclear-armed nations remain consistently short (days to months) for both food security and credit markets. This is the exact mathematical signature of the Malthusian "Last of Us" trap: once output or credit creation is fully predictable, markets systematically low-ball the surplus, collapsing the system on a calculable near-term timeline. The results hold uniformly whether population growth is positive, near-zero, or negative.

These findings confirm that the linear Malthusian form—without any non-linear embellishments—is both necessary and sufficient for policy-grade early-warning in food security and financial stability for nation-states of strategic importance.

The models and the complete SNoG framework (theoretical + empirical) belongs to the Triumvirate of Government, Military, and Central Banking.

References

[1] S. Ghosh, "Food economics in continuous time," 2025.

[2] S. Ghosh, "Predicting food crisis," 2025.

The End